FREEDOM INTERNATIONAL SCHOOL

MATHEMATICS

TOPIC – VECTORS - 1

CLASS: XII **NAME:**

1.	The scalar projection of the vector $3\hat{\imath} - \hat{\jmath} - 2\hat{k}$ on the vector $\hat{\imath} + 2\hat{\jmath} - 3\hat{k}$ is a) $\frac{7}{\sqrt{14}}$ b) $\frac{7}{14}$ c) $\frac{6}{13}$ d) $\frac{7}{2}$	
2.	The projection of $\vec{b} + \vec{c}$ on $\vec{a}$, where $\vec{a} = 2\hat{\imath} - 2\hat{\jmath} + \hat{k}$, $\vec{b} = \hat{\imath} + 2\hat{\jmath} - 2\hat{k}$ and $\vec{c} = 2\hat{\imath} - \hat{\jmath} + 4\hat{k}$ is a) 6 b) 2 c) 1 d) 8	
3.	If two vectors $\vec{a}$ and $\vec{b}$ such that $\lvert\vec{a}\rvert = 2$, $\lvert\vec{b}\rvert = 3$ and $\vec{a}.\vec{b} = 4$, then $\lvert\vec{a} - 2\vec{b}\rvert$ = a) $\sqrt{2}$ b) $2\sqrt{6}$ c) 24 d) $2\sqrt{2}$	
4.	If two vectors $\vec{a}$ and $\vec{b}$ such that $\lvert\vec{a}\rvert = 2\sqrt{2}$, $\lvert\vec{b}\rvert = 2$ and $\vec{a}.\vec{b} = 4$, then the acute angle θ between the vectors $\vec{a}$ and $\vec{b}$ is a) $\frac{\pi}{3}$ b) $\frac{\pi}{4}$ c) $\frac{\pi}{6}$ d) $\frac{3\pi}{4}$	
5.	If $\vec{a}$ and $\vec{b}$ denote the position vectors of point A and B respectively and P is a point on AB such that AP = 2PB, then write the position vector of P. a) $\frac{\vec{a} + \vec{b}}{3}$ b) $\frac{2\vec{a} + \vec{b}}{3}$ c) $\frac{\vec{a} + 2\vec{b}}{3}$ d) $\frac{2\vec{a} + 2\vec{b}}{3}$	
6.	Unit vector in the direction of $\hat{\imath} + 2\hat{\jmath} - 2\hat{k}$ a) $\frac{\hat{\imath} + 2\hat{\jmath} - 2\hat{k}}{3}$ b) $\hat{\imath} + 2\hat{\jmath} - 2\hat{k}$ c) $\frac{-\hat{\imath} - 2\hat{\jmath} - 2\hat{k}}{3}$ d) $\frac{-\hat{\imath} - 2\hat{\jmath} + 2\hat{k}}{3}$	
7.	The cosine of the angle that the vector $\sqrt{2}\hat{\imath} + \hat{\jmath} + \hat{k}$ makes with y-axis is a) $\sqrt{2}$ b) $\frac{1}{2}$ c) 1 d) $-\frac{1}{2}$	
8.	If $\hat{a}$ and $\hat{b}$ are unit vectors and θ is the angle between them, $\lvert\hat{a} + \hat{b}\rvert$ = a) $2\cos\frac{\theta}{2}$ b) $\cos\frac{\theta}{2}$ c) $2\sin\frac{\theta}{2}$ d) $4\cos\frac{\theta}{2}$	
9.	X and Y are two points with position vectors $3\vec{a} + \vec{b}$ and $\vec{a} - 3\vec{b}$ respectively. What will be the position vector of a point Z which divides the line segment XY in the ratio 2:1 externally? a) $\vec{a} - 7\vec{b}$ b) $\vec{a} + 7\vec{b}$ c) $-\vec{a} - 7\vec{b}$ d) $7\vec{a} + \vec{b}$	
10.	If $\vec{a}$ and $\vec{b}$ are two non-zero vectors such that $\lvert\vec{a} \times \vec{b}\rvert = \lvert\vec{a}.\vec{b}\rvert$ then the acute angle between two vectors $\vec{a}$ and $\vec{b}$ is a) $\frac{\pi}{3}$ b) $\frac{\pi}{4}$ c) $\frac{\pi}{6}$ d) $\frac{3\pi}{4}$	

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MATHEMATICS

TOPIC – VECTORS - 2

CLASS: XII **NAME:**

1.	What is the number of vectors of unit length perpendicular to both the vectors $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$? a) 0 b) 1 c) 2 d) 3	
2.	If $\|\vec{a}\| = 2$, $\|\vec{b}\| = 2\sqrt{3}$ and $\vec{a} \perp \vec{b}$, then the value of $\|\vec{a} + \vec{b}\|$ is a) 16 b) 4 c) -4 d) - 16	
3.	Which of the following represents the position vectors of the mid-point of line segment joining the points A $(\hat{i} - 2\hat{j} + 7\hat{k})$ and B $(5\hat{i} + \hat{k})$? a) $3\hat{i} - \hat{j} + 4\hat{k}$ b) $2\hat{i} + \hat{j} - 3\hat{k}$ c) $-3\hat{i} + \hat{j} - 4\hat{k}$ d) $6\hat{i} - 2\hat{j} + 8\hat{k}$	
4.	Projection of vector $\vec{r} = 3\hat{i} - 4\hat{j} + 12\hat{k}$ on x-axis is a) 0 b) 1 c) 3 d) 8	
5.	If $\vec{a}$ and $\vec{b}$ are two perpendicular unit vectors then, the value of $\|\vec{a} + 2\vec{b}\|$ is a) $\pm\sqrt{5}$ b) 5 c) $\sqrt{5}$ d) 3	
6.	Let $\vec{a} = x\hat{i} + 3\hat{j} - 6\hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} - y\hat{k}$. If $\vec{a}$ and $\vec{b}$ are collinear vectors then, value of x + y is a) -6 b) -2 c) -8 d) -3	
7.	A vector with a magnitude of 2 and in the direction of $\hat{i} + 2\hat{j} - 2\hat{k}$ is a) $\frac{2}{3}\hat{i} + \frac{4}{3}\hat{j} - \frac{4}{3}\hat{k}$ b) $2\hat{i} + 4\hat{j} - 4\hat{k}$ c) $-\frac{2}{3}\hat{i} - \frac{4}{3}\hat{j} + \frac{4}{3}\hat{k}$ d) $\pm\left(\frac{2}{3}\hat{i} + \frac{4}{3}\hat{j} - \frac{4}{3}\hat{k}\right)$	
8.	If two vectors $\vec{a}$ and $\vec{b}$ are such that the angle between them is θ and $\|\vec{a}\| = 2$, $\|\vec{b}\| = 3$, $\cos\theta = \frac{2}{3}$ then, $\|\vec{a} + \vec{b}\|$ = a) $\pm\sqrt{21}$ b) 21 c) $\sqrt{24}$ d) $\sqrt{2}$	
9.	Let $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - 4\vec{b}$ are perpendicular. The angle θ between $\vec{a}$ and $\vec{b}$ is a) $\frac{\pi}{3}$ b) $\frac{\pi}{4}$ c) $\frac{\pi}{6}$ d) $\frac{\pi}{2}$	
10.	If θ is the angle between two non-zero vectors $\vec{a}$ and $\vec{b}$ such that $\|\vec{a} + \vec{b}\| = \|\vec{a} - \vec{b}\|$ then θ = a) 0 b) $\frac{\pi}{4}$ c) π d) $\frac{\pi}{2}$	

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TOPIC - VECTORS

CLASS: XII **NAME:**

1. If the vectors $\vec{a}$ and $\vec{b}$ are such that $|\vec{a}| = \frac{1}{2}$, $|\vec{b}| = \frac{4}{\sqrt{3}}$ and $|\vec{a} \times \vec{b}| = \frac{1}{\sqrt{3}}$ then find $\vec{a}.\vec{b}$

2. If $\vec{a}$ and $\vec{b}$ are unit vectors and $\vec{a} + \vec{b}$ is also a unit vector, then what is the angle between $\vec{a}$ and $\vec{b}$?

3. Find the area of the parallelogram whose diagonals are represented by the vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 3\hat{i} + 4\hat{j} - \hat{k}$.

4. a) Find $|\vec{a} - \vec{b}|$, if $|\vec{a}| = 2$, $|\vec{b}| = 5$ and $\vec{a}.\vec{b} = 8$.

   b) If $|\vec{a}| = 13$, $|\vec{b}| = 5$ and $\vec{a}.\vec{b} = 60$, find $|\vec{a} \times \vec{b}|$.

5. Suppose $\vec{a}$, $\vec{b}$ and $\vec{c}$ are vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{c}| = 5$, then evaluate $\vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a}$

6. Find the projection of $\vec{b} + \vec{c}$ on $\vec{a}$ where $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$.

7. Show that $A(\hat{i} - 2\hat{j} - 8\hat{k})$, $B(5\hat{i} - 2\hat{k})$ and $C(11\hat{i} + 3\hat{j} + 7\hat{k})$ are collinear points.

8. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = \hat{j} - \hat{k}$ are given vectors then, find a vector $\vec{b}$ such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a}.\vec{b} = 3$.

9. a) Find the direction cosines and direction ratios of the vector joining the points A(1,2,-2) and B (-1,-2,2) directed from A to B.

   b) The position vectors of the points P, Q, R and S are respectively $\hat{i} - 3\hat{j} + \hat{k}$, $2\hat{i} + \hat{j}$, $3\hat{i} + 2\hat{j} - 3\hat{k}$ and $\hat{i} - 6\hat{j} - \hat{k}$. Prove that the lines PQ and RS are parallel and the ratio of their lengths is 1: 2.

10. a) Find a unit vector perpendicular to each of the vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 3\hat{i} + 4\hat{j} - \hat{k}$.

    Also obtain the sine of the angle between the vectors $\vec{a}$ and $\vec{b}$.

    b) Show that the points A, B, C with position vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$ respectively, are the vectors of a right – angled triangle. Hence find the area of the triangle.

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